Fizika

9-sinf

Mavzu: **Optik asboblar.**










BenQ






HDV
SONY


OLYMPUS



Lupa

Lupa – buyumlarni ko‘rish burchagini kattalashtirib beradigan kichik focus masofali qavariq linza.

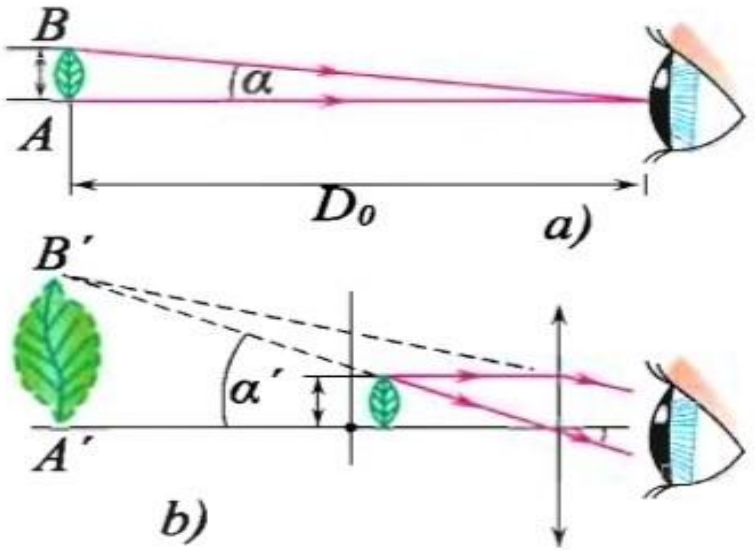


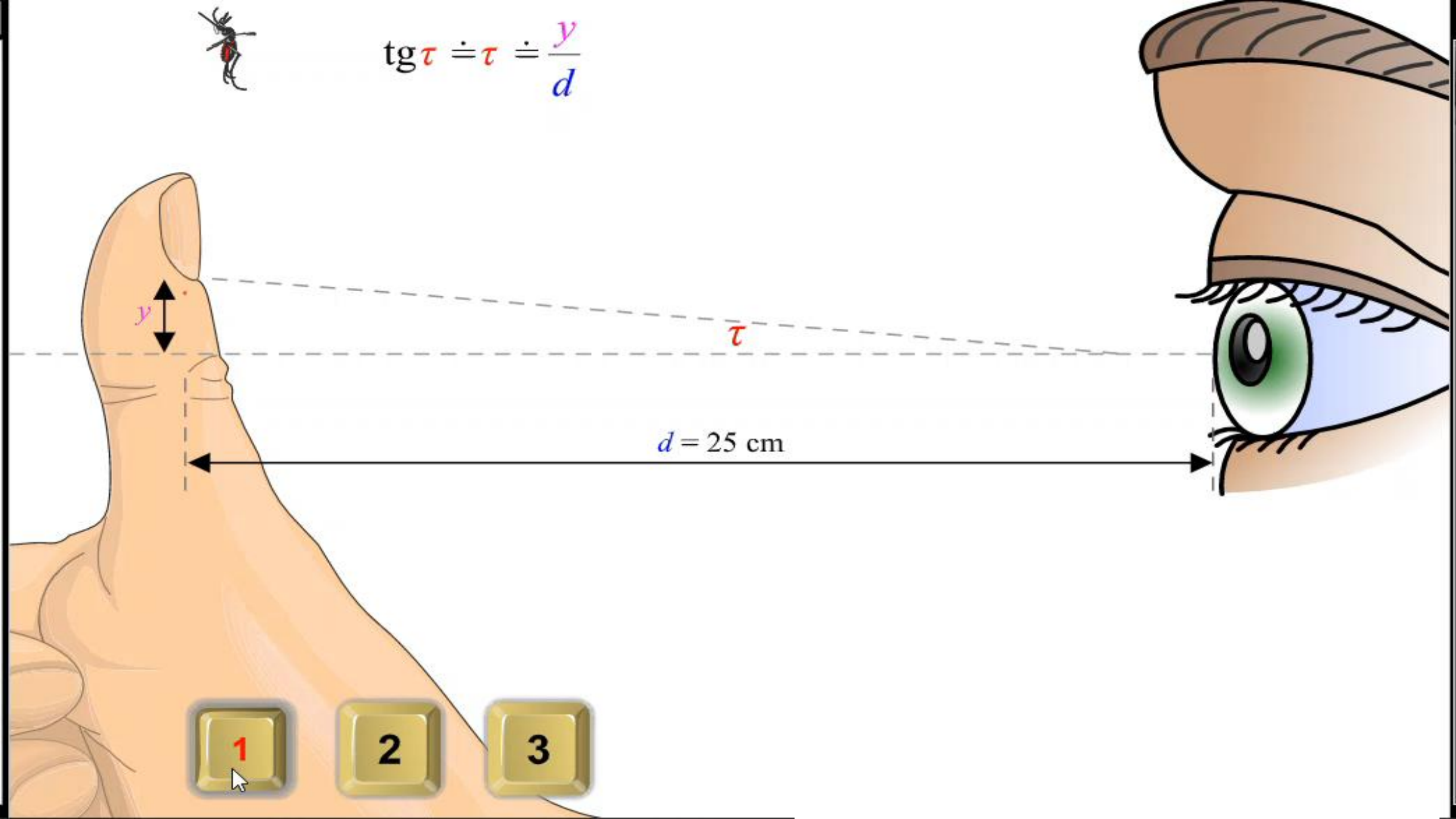
$$D_0 = 25 \text{ cm}$$

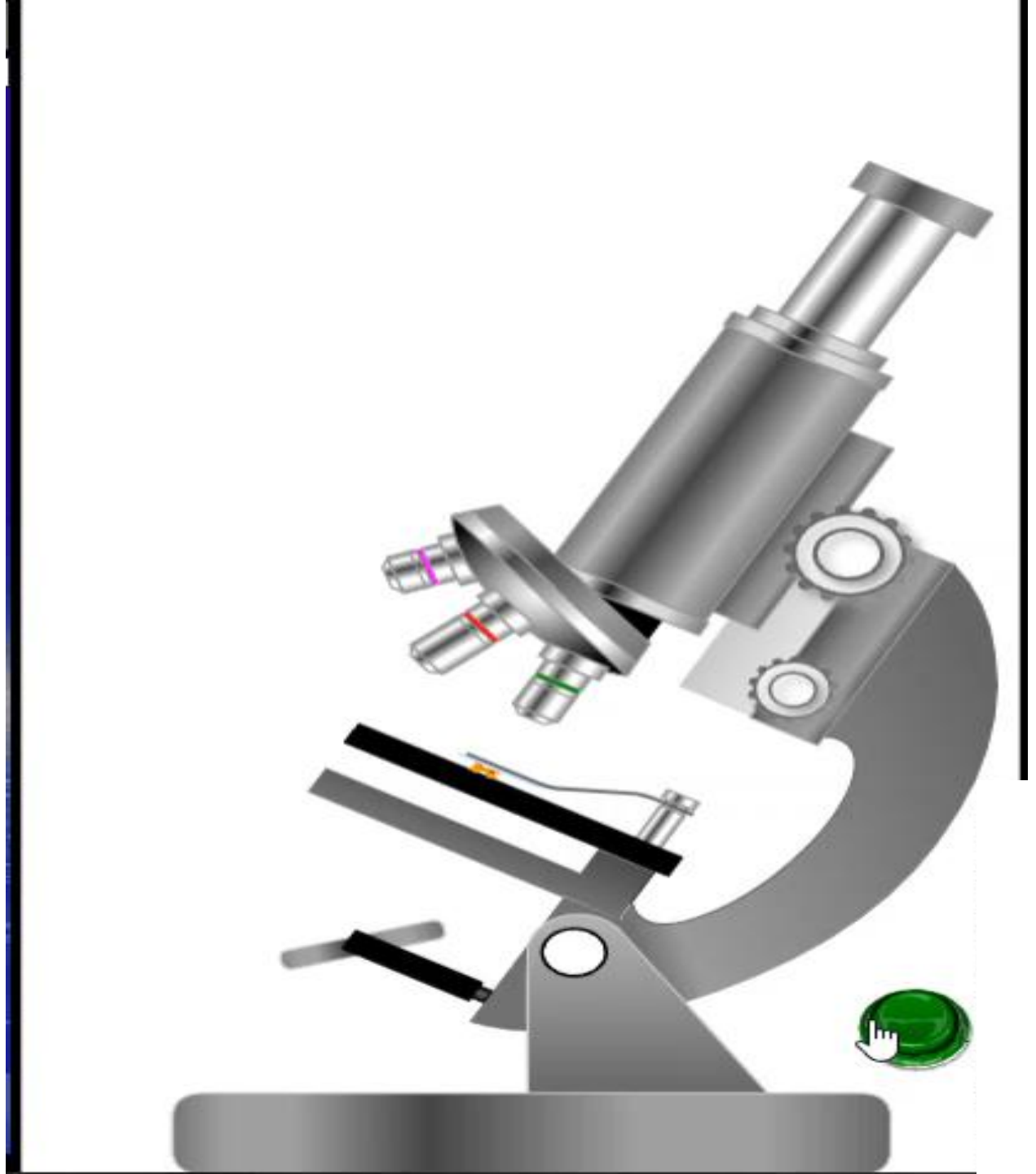
$$K = \frac{A'B'}{AB} = \frac{\alpha'}{\alpha}$$

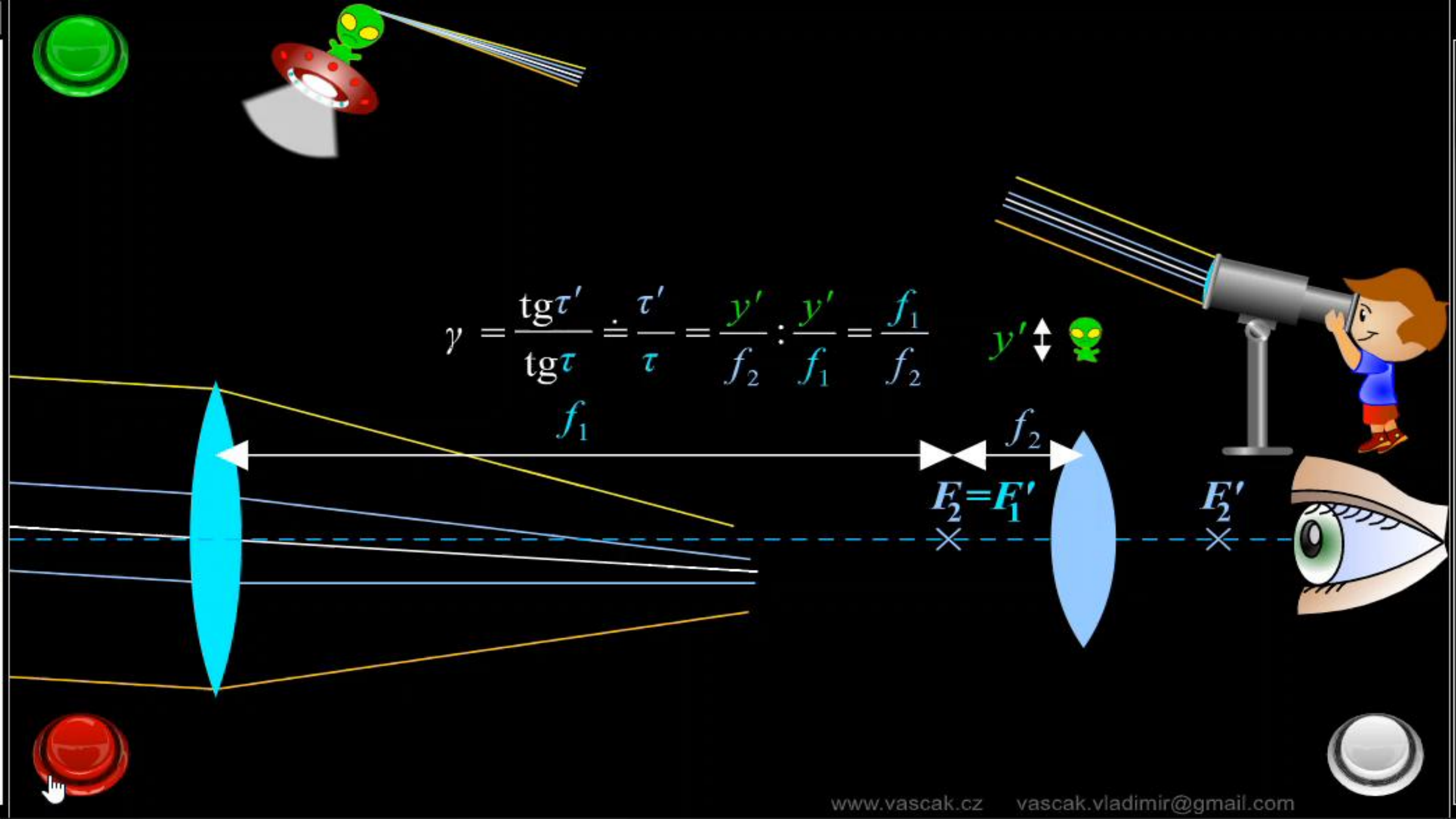
$$K = \frac{D_0}{F}$$

$$F = 1 \div 10\ cm; \quad K = 2{,}5 \div 25.$$



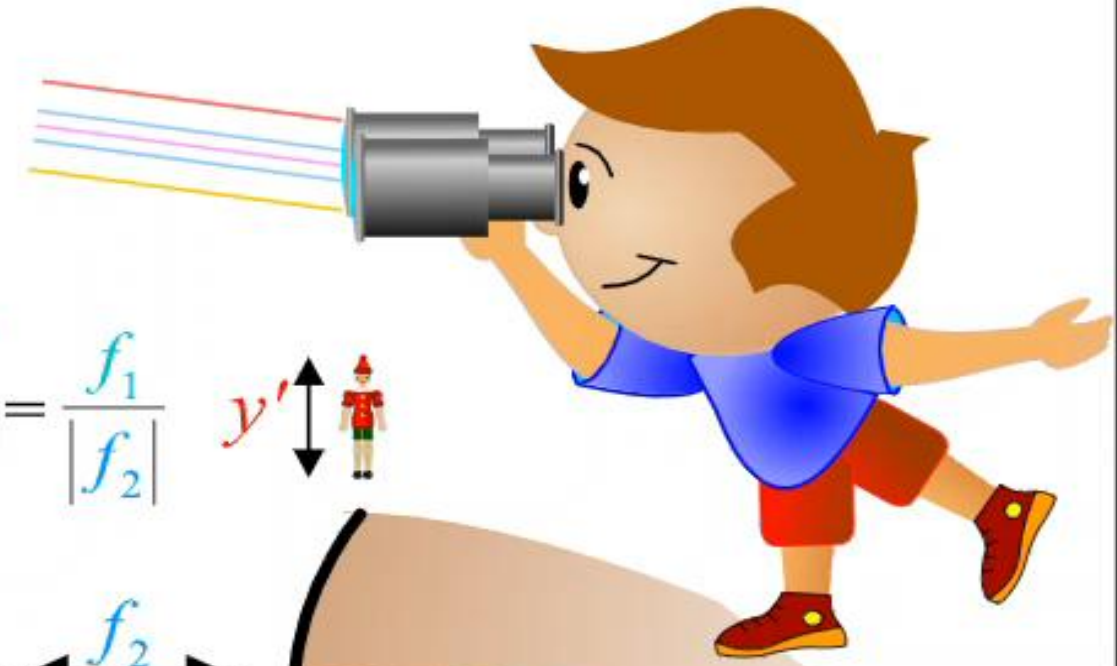

$\mathrm{tg}\,\tau \doteq \tau \doteq \frac{y}{d}$
y
τ
d = 25 cm
1
2
3



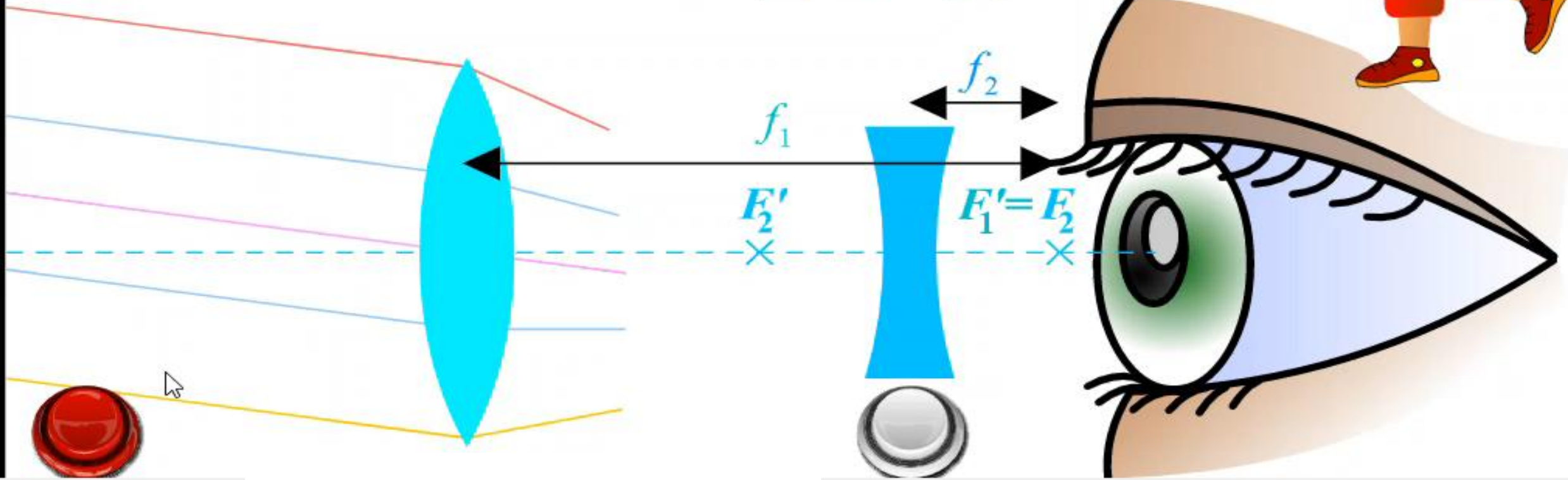


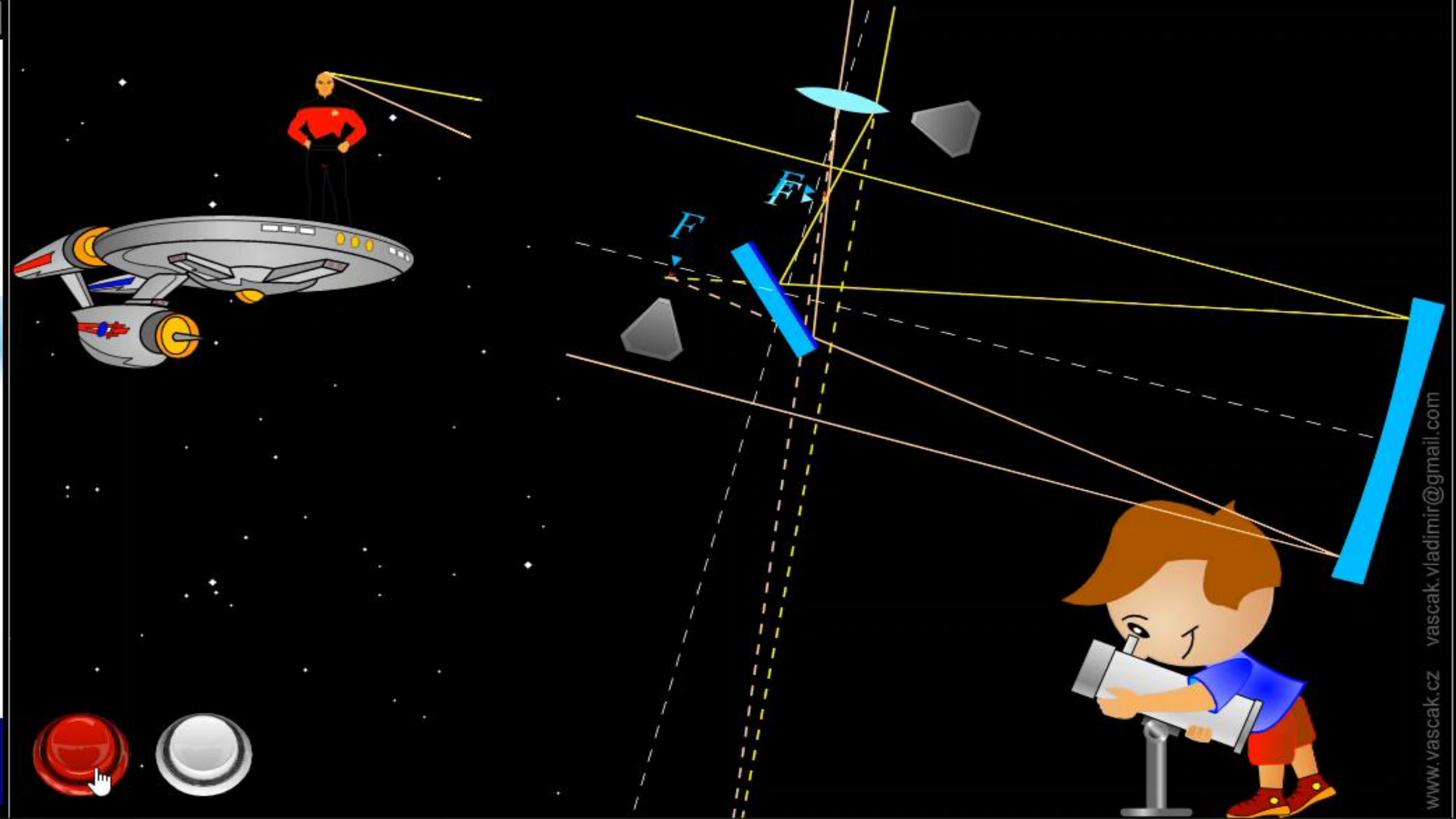
www.vascak.cz vascak.vladimir@gmail.com





$$\gamma = \frac{\operatorname{tg}\tau'}{\operatorname{tg}\tau} \doteq \frac{\tau'}{\tau} = \frac{y'}{|f_2|} : \frac{y'}{f_1} = \frac{f_1}{|f_2|}$$

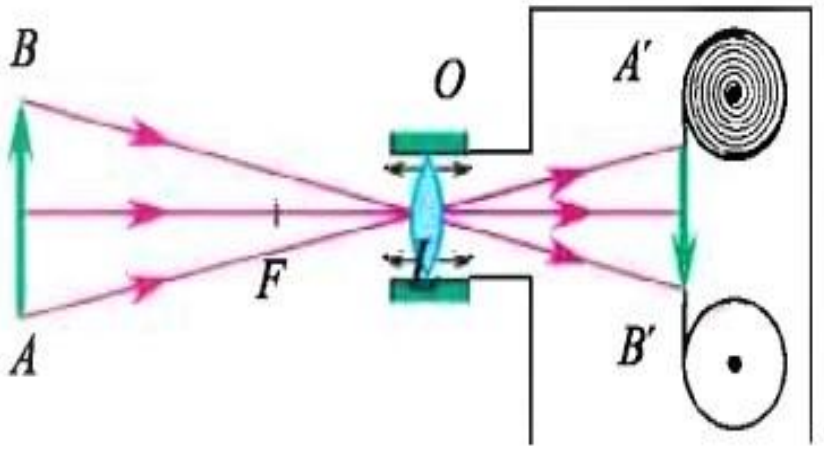



F
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Fotoapparat

Fotoapparat–obyektning tasvirini fotoplyonka, fotoplastina yoki fotoqog‘ozga tushirib, saqlaydigan qilib beradigan asbob.



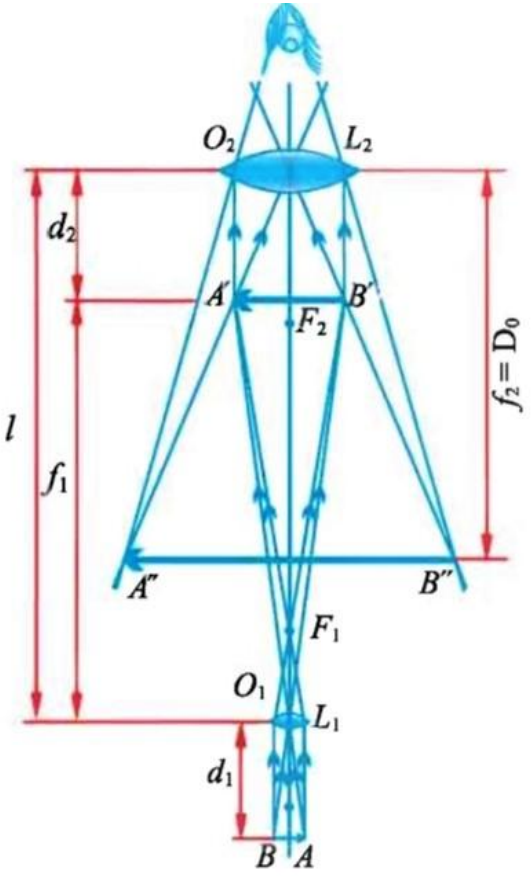


Mikroskop

Mikroskop – yaqin masofadagi ko‘zga bevosita ko‘rinmaydigan juda mayda obyektlarni kattalashtirib ko‘rsatadigan optik asbob.

$$K = \frac{l \cdot D_0}{F_1 \cdot F_2}$$





28-mashq 1-masala

Fokus masofasi 2,5 cm bo‘lgan lupa buyumni necha marta kattalashtirib ko‘rsata oladi? $D_0 = 25\ \text{cm}$

Berilgan:

$F = 2{,}5\ cm = 0{,}025\ m$

$D_0 = 25\ \text{cm} = 0{,}25\ \text{m}$

Topish kerak: $K - ?$

Formula:

$$K = \frac{D_0}{F}$$

Yechish: $K = \frac{0{,}25\ m}{0{,}025\ m} = 10$

Javob: $K = 10$ **(10 marta kattalashtiradi).**

Mustaqil bajarish uchun topshiriqlar

1. Mavzuni o‘qish va savollarga javob yozish. (153-bet)

2. 28-mashqning 2-3-masalasini yechish. (153-bet)